

Mahmoud Abd El-Ghaffar (Mahmoud Zanaty)

AI Agent Architect & Workflow Automation Specialist | Flowise & n8n Engineer

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EXECUTIVE SUMMARY

Accomplished Computer Engineer and AI Agent Architect with extensive hands-on experience designing, extending, and deploying autonomous AI agent networks, enterprise Pinecone RAG pipelines, and complex multi-platform automation systems. Engineered over 30+ production workflows and 10+ native custom Flowise nodes bridging enterprise tools (HubSpot, Lark, ClickUp, Trello, WhatsApp, Telegram, PostgreSQL) with cutting-edge LLMs (OpenAI GPT-4o, Anthropic Claude 3.5 Sonnet, Google Gemini 2.0 Flash).

Utilizes AI-driven development tools (Google Antigravity, Claude Code, OpenAI Codex, VS Code) to rapidly extend low-code platforms (Flowise AI, n8n). Proven capability in engineering 10+ native custom Node.js nodes, implementing Model Context Protocol (MCP) servers, constructing LLM-to-SQL database agents, building custom Webhooks and Schedulers, and embedding PostgreSQL execution loggers for full agent observability and enterprise compliance.

CORE TECHNICAL COMPETENCIES

- Workflow AI: Flowise AI (10+ Custom Nodes Engine), n8n, MCP, Webhooks, Schedulers (+30 Workflows)
- AI Development Tools: Google Antigravity, Claude Code, OpenAI Codex, VS Code AI Suite
- Databases & RAG: Pinecone Vector Database, PostgreSQL (LLM-to-SQL), PGVector, Embeddings
- Integrations: HubSpot CRM, Lark API, ClickUp API, Trello API, WhatsApp API, Telegram, Gmail

TECHNICAL WORK EXPERIENCE

AI & Automation Engineer — MicroMind (Sep 2023 – Apr 2025)

- Applied AI coding tools (Antigravity, Claude Code, Codex) within VS Code to extend Flowise AI and n8n platform capabilities, engineering custom native nodes, custom JS functions, webhooks, and schedulers.
- Architected and deployed over 30+ production AI workflows serving retail, real estate, field engineering, operational management, and content publishing.
- Engineered over 10+ native Flowise integration nodes for WhatsApp, Telegram, Gmail, Weather API, MCP Client, PostgreSQL Logger, Pinecone Handler, Custom Webhooks, Cron Schedulers, and Custom JS Nodes.
- Implemented Model Context Protocol (MCP) agent nodes allowing operations teams to execute backend Node.js scripts dynamically via natural language prompts evaluated by Claude 3.5 Sonnet.
- Engineered natural-language-to-SQL query agents over PostgreSQL project databases for Site Engineers & Field Sales Dashboard managers.
- Built high-speed semantic document retrieval systems using Pinecone Vector Database and OpenAI Embeddings for enterprise RAG applications.
- Constructed a full PostgreSQL Agent Logger capturing every node input, output, tool call, and prompt into a structured audit table with LangSmith telemetry.

Automation Engineer — Operatives (IT Services) (Jun 2021 – Jan 2022)

- Analyzed cross-departmental operational cycles and engineered end-to-end automated workflows, eliminating manual data entry bottlenecks across IT operations teams.

Operation Engineer & Admin — Operatives (IT Services) (Jan 2020 – Jun 2021)

- Managed IT infrastructure operations, monitored workflow execution logs, enforced SLA compliance, and administered operational tracking systems.

DETAILED PROJECT ARCHITECTURE HIGHLIGHTS

- Flowise Multi-Agent Architecture: Hierarchical supervisor & sequential sub-agent routing network.
- Pinecone Enterprise RAG System Design: Ingestion, hybrid vector search, and context audit verification.
- MCP Server Dispatcher Topology: Natural language command interface to local Node.js backend execution.
- Self-Developed Custom Flowise Nodes (10+ Nodes): Native Node.js integration for WhatsApp, Telegram, Gmail, Weather, MCP, PostgreSQL Logger, Pinecone, Schedulers, and Webhooks.
- Site Engineer Assistant: Secure LLM-to-SQL query generation over PostgreSQL construction databases.
- Field Sales Dashboard Agent: Van-level performance & target KPI query engine with natural language output visualization.
- Retail Support & Installment Eligibility AI: Integrated PostgreSQL, Pinecone, HubSpot CRM, and Gmail follow-up engine.

- Real Estate Sales Assistant: Multi-modal listing search, dynamic recommendations, and instant HubSpot CRM synchronization.
- Chat-Triggered Trello Task Creator: Conversational AI node interpreting team messages into Trello cards.
- AI News & Social Publisher Bot: n8n workflow connecting NewsAPI -> Google Sheets -> Facebook Graph API.
- PostgreSQL Agent Audit Logger & LangSmith: Complete execution audit system logging inputs, outputs, tools, and prompts.

EDUCATION

B.Sc. in Computer Engineering — Faculty of Engineering Shoubra, Benha University

Location: Benha, Egypt | Specialization: Computer Systems Architecture, Database Management Systems, Software Engineering & AI Systems.